VU Medical Zone

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ZOO 503

Lecture no 01

1-Palentology is the study of fossils to determine organisms…….. Evolution & ecology.

2-The word paleontology originates from Greek “palaios…”, “ontos…." “Logos…."means the …… old, creature, study.

3-Paleontology lies on the border between…….. Biology & geology.

4- The Earth went through a phase of substantial global warming, a drop in oxygen levels and acid rain about the…….. 250 million years ago.

5-The use of fossils in dating rocks, is a powerful tool for understanding deep time, and it is widely used in scientific studies this branch is known as……..Biostratigraphy.

LECTURE NO 02

1- From Latin scientia, meaning……."Knowledge”.

2-In natural sciences, there have been two main approaches……. Induction & deduction.

3-A series of observations point to an inevitable Outcome“All men are mortal. Socrates is a man. Therefore Socrates is mortal.’’……..Aristotle.

4-The Earth had been hit by a meteorite …….. Years ago, caused the extinction of the…….. 65 million years ago, dinosaurs.

LECTURE NO 03

1- Fossil sharks’ teeth were known as …….worn as amulets……(*glossopteris).*
2- One of the first to hint at the idea of extinction…….Robert Hooke.

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3- Father of comparative anatomy……..Cuvier.

LECTURE NO 04

1- Limited Knowledge of the fossil record in the …….. 1820s, 1830s.
2- Charles Lyell (1797–1875), was an…… antiprogressionist.
3- "Great Chain of Being" or the Scala naturae."All organisms were linked in time by a unidirectional ladder, from simplest at the bottom to most complexes at the top" is the idea of……..Lamarck's.
4- Charles Darwin (1809–1882), developed the theory of evolution by natural selection in the…….1830s.
5- "Modern synthesis" Modern genetics early in the…….20th century.
6- Book "On the Origin of Species''…… 1859.

LECTURE NO 05

1- 19th century paleontology: "Archaeopteryx", true…..'' is missing link''.
2- In 1947, the first soft-bodied Ediacaran fossils were found in……Australia.

LECTURE NO 06

1- Fossils information is also useful in…….Geology.

LECTURE NO 07

1-How many types of the stratigraphy…….8

LECTURE NO 08

1- Earth as a living organism in equilibrium with its biosphere…….Gaia hypothesis.
2- Earth as a dynamic and changing system…….Modern Concept.
3- The order and succession of rocks is called…….Lithostratigraphy.
4- A succession of contiguous formations with common characteristics is called a…….group.

LECTURE NO 09

1-"Assemblages of fossils in different layers" work of ……..William Smith in Britain.

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2-Paleozoic (“ancient life”) Mesozoic (“middle life”) Cenozoic (“recent life”) work of ……John Phillips.

3-The main operational units of a biostratigraphy……Biozones.

LECTURE NO 10

1-“Since the fossil record of organisms is never complete, neither the first nor the last organism in a given taxon will be recorded as a fossil” effect is known as the ……Signor–Lipps effect.

2-Graptolites (1 myr) and Ammonites (25 kyr) are the best known and most reliable zone……..Macrofossil.

3-“The branch of geology concerned with establishing the absolute ages of strata.” Is called the……..Chronostratigraphy.

4-The study of astronomically forced climate cycles within sedimentary successions is called……..Cyclo stratigraphy.

LECTURE NO 11

1-The Earth today can be divided into main provinces……..six.

2-The emergence of the Isthmus of Panama……. (3 Mya).

LECTURE NO 12

1-The study of the species composition and species richness on islands is called……Island biogeography.

2-The cross-latitude transfer of evolving animals and plants may have acted as…… “Noah’s arks”.

3-Galápagos, or Aldabra, have become important sites for biologists to watch…… Evolution in action.

LECTURE NO 13

1-An eminent Alpine geologist……..Rudolf Trümpy.

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2-The wide-hinged fossils are……. “Spiriferide brachiopods”.

LECTURE NO 14

1-The paleontological study includes all the processes that occur after the death of an organism and before its fossilization in the rock is called……Taphonomy.

LECTURE NO 15

1-How many kinds of the fossils…….2.

2-The partial or complete remains of plants or animals……Body fossils.

3-The remains of the activity of ancient organisms, such as burrows and tracks……Trace fossils.

3-When large animals feed on dead plant or animal tissues, the process are termed…….scavenging.

4-When microbes, such as fungi or bacteria, transform tissues of the dead organism, the process is termed……decay.

5-Soft parts of animals are made from ……volatiles.

6-Refractories are the organic carbons much less liable to break down, such as…….cellulose.

7-Physical and chemical effects occurring after burial are termed as……Diagenesis.

8-Mineralization of soft tissues occurs in ways……..3.

LECTURE NO 16

1-Skeletons that are made from several parts may become disarticulated, separated into their component parts. After scavenging or decay of connective tissue is called…….Disarticulation.

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2-The removal of skeletal materials by boring organisms such as sponges, algae and bivalves is called the……..Bio erosion.

3-Addition of CaCO3 is called…….. Carbonate concretion.

LECTURE NO 17

1- "Imperfection of the geological record'' statement of ……..Charles Darwin.

LECTURE NO 18

1-The functions of single fossil organisms……pale autecology.

2-The composition and structure of fossil communities……pale synecology.

3-Climate has been an important factor in driving change at a number of different levels……Evolutionary changes.

LECTURE NO 19

1-The similarity of a death assemblage to its living counterpart……fidelity.

2-Taphocoenosis is the …… product that is finally preserved…….End.

LECTURE NO 20

1-The building blocks of the communities are called the …….populations.

2-A naturally occurring assemblage of plants and animals that live in the same place at the same time and regularly interbreed is called ……..population.

3-"All the populations of species living in association" is called…….ecosystem.

4-Species can occupy the same ecological niche for many millions of years, adding stability to many ecosystems are called the…….Incumbent Species.

5-Fossil organism can be defined in the term of…… habitat& niche.

5- The organism lives beneath the sediment–water interface are called the…….In faunal organisms.

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6- The organisms live above it the sediment–water interface is called……Epifaunal organisms.

7- The groups of functionally similar organisms occurring together in a community are called the…….Guilds.

LECTURE NO 21

1-Last 600 million years Earth has oscillated at least times between icehouse and greenhouse conditions……..five.

2-There are how many numbers of the climatic zone……5.

3-Soils rich in calcium carbonate is called…..calcretes.

4-Stones that plummet from the bottoms of melting icebergs into seabed sediment are known as……Dropstones.

LECTURE NO 22

1-Evolution by natural selection is a core scientific model that was set out by……Darwin.

2-Speciation often occurs by the establishment of a …….. Barrier.

3-The evolution of life may be represented by a single branching…….. phylogenetic tree.

3- A method of reconstructing phylogeny based on the identification of shared derived characters (homologies) Is called the ……..Cladistics.

LECTURE NO 23

1-Darwin laid the framework for Evolutionary Biology ……150 years ago.

2-Paleontological aspects of evolution, such as the tree of life and studies of processes over thousands and millions of years, are sometimes called……macroevolution.

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3-“Only the organisms best adapted to their environment tend to survive and transmit their genetic characteristics in increasing numbers to succeeding generations while those less adapted tend to be eliminated.” Is called the ……principle of the natural selection.

4-Nearly all species produce far more young than can survive to adulthood is known as the principle of the……Malthus’ principle.

LECTURE NO 24

1-The cohesion of a species is maintained over its natural range by processes of……Gene flow.

2-The process of splitting of a population to form two species is called……. Speciation.

LECTURE NO 25

1-Ladder go from lower to higher, simple to ……complex.

2-Phylogenetically uninformative characters fall into two main categories……Convergences, Plesiomorphies.

3-Clades are distinguished from two kinds of non-natural groups……Paraphyletic groups, Polyphyletic groups.

LECTURE NO 26

1-Apomorphies are features that arose once only in…… evolution.

2-The phylogenetic ally informative character is called……..Henning.

3-“Apomorphies” shared by two or more species are termed……Synapomorphies.

LECTURE NO 29

1-The ratio of eye diameter to body length diminishes as the animal approaches adulthood is an example of the…… Allometric growth.

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2-If there is no change in proportions during growth, the feature is said to show…….. Isometric growth.

LECTURE NO 30

1-In 1866, Ernst Haeckel, a German evolutionist, a law that is “ontogeny recapitulates phylogeny”. Known as the……Biogenetic Law.

2-Certain developmental abnormalities called……..atavisms.

LECTURE NO 31

1-During mass extinctions of species were wiped out; these include abroad range of organisms……..20-90%.

2-How much Phanerozoic mass extinction occurred ……..5.

LECTURE NO 32

1-The dodo is perhaps the most iconic of……Icons.

2-The most spectacular extinctions are known as……mass extinctions.

LECTURE NO 33

1-Paleontologists talk about the “big five” mass extinctions of the last ……..540 myr.

2-The Phanerozoic, and the current extinction crisis is sometimes called the…….. “Sixth extinction”.

3-Mass extinctions through the past ……..600 myr.

4-The idea of periodicity of impacts was reawakened by Rohde and Muller (2005) who argued for a periodicity in mass extinction…….62 myr.

LECTURE NO 35

1-A marine mass extinction event at, or close to, the Triassic-Jurassic boundary…… 200 Ma.

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2-1991, Crater was identified in Chicxulub in…… Mexico.

3-The Deccan Traps in India represent a vast outpouring of lava that occurred over the spanning the KT boundary……..2-3myr.

LECTURE NO 36

1-Photosynthesis by a group of bacteria, called…… Cynobacteria.

2-Bacteria in rocks up to 3.2 Ga, indicated by ……..stromatolites.

3-The first widely accepted cellular fossils date from ……2.5 Ga.

4-The oldest eukaryotes…… 1.9 Ga.

5-Red algae from show that sex had originated……1.2 GA.

LECTURE NO 37

1-Simple organic molecules in meteorites called…… carbonaceous chondrites.

2-The biochemical theory for the origin of life, as proposed by Oparin and Haldane in the ……..1920s.

3-The last universal common ancestor of life (LUCA) was a hyper thermopile that lived in ……hot water conditions.

LECTURE NO 39

1-The transition from non-living to living is called the ……..precellular life.

2-RNA, or ribonucleic acid, is one of the nucleic acids and it has key roles in…… protein synthesis.

3-Walter Gilbert first used the term “RNA world” in…… 1986.

4- RNA could act as a precursor of…… Life.
5- An RNA that could edit out unnecessary parts of the message it carried before delivering it to the ribosome called his discovery a ……ribozyme.

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6- A membrane-bound structure composed mainly of lipids that self-replicates called the ……protocell.

LECTURE NO 40

1-The oldest rocks are from Canada and are dated at ……3.8–4 Ga.

2-The oldest sedimentary rocks from the Isua Group in Greenland dated at ……..3.8–3.7 Ga.

3-Around 0.8–0.6 Ga is indicated by increased levels of ……marine sulfate.

LECTURE NO 41

1-Endosymbiotic theory was proposed by the Margulis in……1970s.

2-The oldest eukaryote is controversial Lipid biomarkers indicate that eukaryotes were around at least by…… 2.7 Ga.

3-The oldest eukaryote fossil may be Grypania, a coiled, spaghetti-like organism that has been reported from rocks as old as ……1.85 Ga.

4-One of the oldest multicellular organisms/eukaryote is ……..Bangiomorpha 1.2 Ga.

LECTURE NO 42

1-Single-celled animal-like protozoan are called the ……..foraminifera.

2-plant-like protozoans with silica skeletons known as the……diatoms.

3-Animla like protozoans with siliceous tests known as the……Radiolarians.

LECTURE NO 44

1-Neither animal nor plant, but single-celled eukaryotes that show animal characteristics such as motility and heterotrophy is known as the …….protozoa.

2-Dinoflagellate biomarkers have been identified in Upper Proterozoic and ……Cambrian rocks.

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2- Consist of some 8000 species of single-celled organisms that swim by beating cilia is called the ……..ciliophora.

LECTURE NO 45

1-A paraphyletic group of eukaryotes that usually contains chloroplasts with chlorophyll c is called the…….Chromistans.

2-The plankton less than 63 µm known as the……Nano plankton.

LECTURE NO 46

1-Unicellular Autotrophs that are included among the Chrysophyte algae characterized by large green-brown chloroplasts are known as the ……diatoms.

2-Prefer marine environments are called the ……The Centrales.

3-More common in freshwater lake are called the……The Pennales.

LECTURE NO 47

1-Spiralians and Ecdysozoans are known as the…….the protostomes.

2-Metazoans originated before Ediacaran…….. 600 Ma.

LECTURE NO 48

1-Molecular data suggest metazoans have been around for at least ……. 600 Myr

2-Earliest metazoans occur in Ediacara biota ……600–550 Ma.

3-The oldest locomotory trace fossils are from about ……550 Ma northwest Russia.

4-Biomarkers, essentially the biochemical fingerprints of ……life.

LECTURE NO 49

1-Body plans are usually defined by the number and type of enveloping walls of tissue together with the presence or absence of a ……Ceolom.

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2-In cnidarians two layers are separated by acellular……..mesogloea.

LECTURE NO 50

1-. “First mouth” mouth develops directly from the blast pore in the …… protostomes.

2-‘’Second mouth” however, have a mouth arising from a secondary opening in the …….deutrotomes.

LECTURE NO 51

1-The three great evolutionary faunas of Phanerozoic, Cambrian, and Paleozoic modern Developed during last…….. 550 myr.

LECTURE NO 52

1-The majority of “Paleozoic” taxa were derived from ……..Cambrian stocks.

LECTURE NO 53

1-Out of 25 or so commonly recognized animal phyla fewer than (35%) have an adequate fossil record…….9(Nine).

2-ThePlatyhelminthes or the flatworms are ……..bilateral animals.

3- A diverse polychaete fauna has been described from the…… Burgess Shale.

LECTURE NO 54

1- One hypothesis suggests that a Precambrian colonial organism may have been a source for the ……bilaterians.

LECTURE NO 55

1- “Porifera is a unique group in its own right”. ……Dr Robert Grant.
2- Sacs with a single chamber lined by flagellate cells called the……ascon.
3- A number of simple chambers with a single central paragaster…….Sycon.
4- Most common where a series of sycon chambers access a large central paragaster……leucon.

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LECTURE NO 56

1-The cnidarians has two life forms……..medusa (swim), polyp (sessile).

LECTURE NO 57

1- A filamentous feeding organ is called……Lophophores.
2- Stenolaemata is a class of marine ……..bryozoans.
3- All extant species are in the order ……Cyclostomatida.

LECTURE NO 58

1- Most successful invertebrate phyla….. ……`Brachiopoda.

LECTURE NO 59

1- The brachiopods experienced main extinction events……5.

LECTURE NO 60

1- A ring of tentacles direct food to mouth leading into a U shaped gut……..10.
1- Stenolaemata dominated Paleozoic bryozoans…… Faunas.

LECTURE NO 61

1- Annelid worms were a sister group to the…… mollusks.

LECTURE NO 62

1-The radula of the mollusks is composed of the …….chitin.

LECTURE NO 63

1- A group of shell-less mollusks…….Aplacophorans.
2- Possesses a radula, a broad foot and a stiffened dorsum, so placing it firmly within the mollusks, close to Kimberella called the……Odontogriphus.

LECTURE NO 64

1- Bivalves are twin-valved shellfish superficially resembling the …… brachiopods.

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LECTURE NO 65

1- The "belly-footed" mollusks are called the ……..gastropods.
2- Most Paleozoic gastropods are herbivores or ……detritus feeders.

LECTURE NO 66

1- Most highly organized of the mollusks are called…….cephalopods.
2- The cephalopodas with straight or coiled external shells with simple sutures……Nautiloidea.
3- The cephalopodas with coiled, commonly ribbed external shells with complex structures……Ammonoidea.
4- The cephalopodas with straight or coiled internal skeletons…….Coleo idea.

LECTURE NO 67

1- The first scaphopods appeared during the and apparently had similar lifestyles to living forms such as ……..Devonian, Dentalium.

LECTURE NO 68

1- The gastropods and cephalopods were probably derived independently from a separate…… monoplacophoran ancestor.

LECTURE NO 70

1- The first major faunas: Early Cambrian……Ecdysozo.
2- Earliest terrestrial body fossils in the Mid Ordovician are ……Myriapods.
3- There are probably million species of living insects……10.
4- Insects first appeared during the ……Early Devonian.

LECTURE NO 71

1- Arthropods exoskeleton is Hardened or sclerotized by…… $CaCO_3$ or $Ca_3(PO_4)_2$.

LECTURE NO 72

1- Earliest arthropods were ……trilobites.

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2- Trilobite Exoskeleton is divided into ……three lobes.

LECTURE NO 73

1- Diverse and heterogeneous. Mites, scorpions and spiders are known as the……Chelicerates.
2- Prosoma (head and thorax) with …segments with appendages…….6.
3- An opisthosoma (abdomen) with …segments, pair of chelicerae (pincers) on the first segment of the prosoma……..12.
4- The chelicerates including the aquatic horseshoe crab Limulus sand the giant sea scorpions or eurypterids is called the…… Merostomes.
5- The chelicerates mainly comprise the terrestrial spiders and scorpions are called the……arachnids.

LECTURE NO 74

1- The first animals to colonize the land are called…….Myriapods.
2- The giant Arthropleura nearly how meter long……2.
3- An enigmatic group of arthropods with antennae and mandibles is called……Euthycarcinoids.

LECTURE NO 75

1-Hexapoda Can be divided into pterygotes and apterygotes …… with wings, without wings.

LECTURE NO 76

1-Ostracodes are considered part of the……Micro fauna.

2-Ostracodes are crustacean arthropods, abundant and widespread in…… aquatic environments.

LECTURE NO 77

1-The echinoids were part of the mobile ……benthos.

2-Asteroids (starfish) were more important in ……post-Paleozoic rocks.

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3-Graptolites are hemichordates closely related to the living…… rhabdopleurids.

LECTURE NO 78

1-One of the most abundant marine animal groups……..deutrotomes.

LECTURE NO 80

1-A group of phosphate minerals, usually referring to hydroxyl apatite, fluorapatite and chlorapatite, with high concentrations of OH−, F−and Cl−ions, respectively, in the crystal is called the ……..Apatite.

2-The oldest vertebrates are small fish-like creatures from the Early Cambrian…… of China.

3-Conodonts commonly occur as tooth-like elements that are useful in ……..biostratigraphy.

4-Synapsids dominated ecosystems on land during the…… Permian and Triassic.

LECTURE NO 81

1-A network of collagen fibers on which needle-like crystals of hydroxyl apatite (a form of apatite, CaPO4) accumulate is called……bone.

2-Flexible tissue, usually unmineralized, contains collagen and elastic tissues are called……cartilage.

3-jawless, generally armored, had their heyday in the Devonian these fishes are called……Ostracoderms.

4-Extinct genus of primitive jawless fish, closely related to Cephalaspis, that lived in the Devonian period…….Hemicyclaspis.

LECTURE NO 82

1-Animals of mystery are called…….Conodonts.

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LECTURE NO 83

1-The basal vertebrates, lacked jaws, jaws probably evolved during the…….. Ordovician.

2-Oldest jaw-bearing fishes were the placoderms, such as……..Coccosteus.

3-The first shark like chondrichthyans or cartilaginous fishes, came on the scene during the……. Early Devonian.

4-Small fishes, mostly in the range mm in length……..50-200.

LECTURE NO 84

1-Tetrapods arose from fishes during the ……..Devonian.

LECTURE NO 85

1-Enclose the embryo the membrane is called……amnion.

2-Collect waste the membrane is called……allantois.

3-Line the eggshell the membrane is called …….chorion.

4-Late Carboniferous Three main clades distinguished by the pattern of openings in the side of the skull……amniotes.

5-Theanapsid (“no arch”) skull pattern, no ……temporal openings.

6-The synapsid (“same arch”), lower…… temporal opening.

7-The diapsid (“two arch”) pattern, two temporal openings.

7- The three major skull patterns seen in amniotes…… anapsid, diapsid and synapsid.

LECTURE NO 86

1-“mammal-like reptiles” and the mammals are called the…….the Synapsida.

2-All modern amniotes produce the ……cleidoic egg.

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3-The turtles appeared first in the…….. Late Triassic.

LECTURE NO 87

1-Reptiles typically have six bones in the…… lower jaw.

2-Dinosaurs famously ruled the Earth for 160 Myr of the ……Mesozoic.

LECTURE NO 88

1-Dinosaurs were a hugely successful group for160 myr of the……. Mesozoic.

2-The key Mesozoic flyers…….Pterosaur.

3-Most important marine reptiles……. Plesiosaurs and ichthyosaurs.

4-Birds evolved from dinosaurs, and radiated during……Tertiary.

5-Molecular and paleontological evidence show that modern mammals radiated in the ……Late Jurassic and Early Cretaceous.

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